dag_args.jpg

```
# --- Task 1.1: Define DAG arguments ---
default_args = {
    'owner': 'Your Name',
    'start_date': days_ago(0),
    'email': ['your_email@mail.com'],
    'email_on_failure': True,
    'email_on_retry': True,
    'retries': 1,
    'retry_delay': timedelta(minutes=5),
}
```

dag_definition.jpg

```
# --- Task 1.2: Define the DAG ---
dag = DAG(
    'ETL_toll_data',
    default_args=default_args,
    description='Apache Airflow Final Assignment',
    schedule_interval=timedelta(days=1),
)
```

unzip_data.jpg

```
# --- Task 2.1: Unzip data ---
unzip_data = BashOperator(
    task_id='unzip_data',
    bash_command='tar -xzf /home/project/airflow/dags/finalassignment/tolldata.tgz -C
/home/project/airflow/dags/finalassignment',
    dag=dag,
)
```

extract_data_from_csv.jpg

```
# --- Task 2.2: Extract from CSV ---
# Fields: Rowid (1), Timestamp (2), Anonymized Vehicle number (3), Vehicle type (4)
extract_data_from_csv = BashOperator(
    task_id='extract_data_from_csv',
    bash_command='cut -d"," -f1,2,3,4 /home/project/airflow/dags/finalassignment/vehicle-data.csv >
/home/project/airflow/dags/finalassignment/csv_data.csv',
    dag=dag,
)
```

extract_data_from_tsv.jpg

```
# --- Task 2.3: Extract from TSV ---
# Fields: Number of axles (5), Tollplaza id (6), Tollplaza code (7)
extract_data_from_tsv = BashOperator(
    task_id='extract_data_from_tsv',
    bash_command='cut -f5,6,7 /home/project/airflow/dags/finalassignment/tollplaza-data.tsv | tr "\\t" "," >
/home/project/airflow/dags/finalassignment/tsv_data.csv',
    dag=dag,
)
```

extract_data_from_fixed_width.jpg

```
# --- Task 2.4: Extract from Fixed Width ---
# Fields: Type of Payment code (last two fields in the payment-data.txt)
extract_data_from_fixed_width = BashOperator(
    task_id='extract_data_from_fixed_width',
    bash_command='cut -c59-67 /home/project/airflow/dags/finalassignment/payment-data.txt | tr "
" "," > /home/project/airflow/dags/finalassignment/fixed_width_data.csv',
    dag=dag,
)
```

consolidate_data.jpg

```
# --- Task 2.5: Consolidate data ---
consolidate_data = BashOperator(
    task_id='consolidate_data',
    bash_command='paste -d"," /home/project/airflow/dags/finalassignment/csv_data.csv
/home/project/airflow/dags/finalassignment/tsv_data.csv
/home/project/airflow/dags/finalassignment/fixed_width_data.csv >
/home/project/airflow/dags/finalassignment/extracted_data.csv',
    dag=dag,
)
```

transform.jpg

```
# --- Task 2.6: Transform data ---
# Convert vehicle_type (column 4) to uppercase
transform_data = BashOperator(
    task_id='transform_data',
    bash_command='awk -F"," \'BEGIN {OFS=","} {$4=toupper($4); print}\'
/home/project/airflow/dags/finalassignment/extracted_data.csv >
/home/project/airflow/dags/finalassignment/transformed_data.csv',
    dag=dag,
)
```

task_pipeline.jpg

```
# --- Task 2.7: Define the task pipeline ---
unzip_data >> extract_data_from_csv >> extract_data_from_tsv >>
extract_data_from_fixed_width >> consolidate_data >> transform_data
```

submit_dag.jpg

```
theia@theiadocker-sugiantoliau:/home/project$ airflow dags list | grep ETL_toll_data
ETL_toll_data                                      | /home/project/airflow/dags/ETL_t
oll_data.py                                                                          | Y
our Name | True
```

unpause_trigger_dag.jpg

```
[2026-03-19T16:40:34.581+0000] {dagbag.py:545} INFO - Filling up the DagBag from /home/pro
ject/airflow/dags
dag_id        | is_paused
==============+==========
ETL_toll_data | True
```

dag_tasks.jpg

```
n [database] - the old setting has been used, but please update your config.
dag_id                              | fileloc                           | owners    | is_paused
====================================+===================================+===========+==========
ETL_toll_data                       | /home/project/airflow/dags/ETL_   | Your Name | False
                                    | toll_data.py                      |           |
conditional_dataset_and_time_bas    | /home/airflow/.local/lib/python   | airflow   | True
ed_timetable                        | 3.9/site-packages/airflow/examp   |           |
                                    | le_dags/example_datasets.py       |           |
consume_1_and_2_with_dataset_exp    | /home/airflow/.local/lib/python   | airflow   | True
ressions                            | 3.9/site-packages/airflow/examp   |           |
                                    | le_dags/example_datasets.py       |           |
consume_1_or_2_with_dataset_expr    | /home/airflow/.local/lib/python   | airflow   | True
essions                             | 3.9/site-packages/airflow/examp   |           |
                                    | le_dags/example_datasets.py       |           |
consume_1_or_both_2_and_3_with_d    | /home/airflow/.local/lib/python   | airflow   | True
ataset_expressions                  | 3.9/site-packages/airflow/examp   |           |
                                    | le_dags/example_datasets.py       |           |
dataset_consumes_1                  | /home/airflow/.local/lib/python   | airflow   | True
                                    | 3.9/site-packages/airflow/examp   |           |
                                    | le_dags/example_datasets.py       |           |
dataset_consumes_1_and_2            | /home/airflow/.local/lib/python   | airflow   | True
                                    | 3.9/site-packages/airflow/examp   |           |
                                    | le_dags/example_datasets.py       |           |
dataset_consumes_1_never_schedul    | /home/airflow/.local/lib/python   | airflow   | True
ed                                  | 3.9/site-packages/airflow/examp   |           |
                                    | le_dags/example_datasets.py       |           |
dataset_consumes_unknown_never_s    | /home/airflow/.local/lib/python   | airflow   | True
cheduled                            | 3.9/site-packages/airflow/examp   |           |
                                    | le_dags/example_datasets.py       |           |
dataset_produces_1                  | /home/airflow/.local/lib/python   | airflow   | True
```

dag_runs.jpg

```
[2026-03-19T16:42:44.277+0000] {__init__.py:43} INFO - Loaded API auth backend: airflow.ap
i.auth.backend.basic_auth
[2026-03-19T16:42:44.280+0000] {__init__.py:43} INFO - Loaded API auth backend: airflow.ap
i.auth.backend.session
     |        |        |        |        |        | last |        |      |        |
     |        |        |        |        |        | _sch |        |      |        |
     |        |        | data_  | data_  |        | edul |        |      |        |
     |        |        | inter  | inter  |        | ing_ | logic  |      |        |
     | dag_i  | dag_r  | val_s  | val_e  | end_d  | deci | al_da  | run_ | start  |
conf | d      | un_id  | tart   | nd     | ate    | sion | te     | type | _date  | state
=====+=======+=======+=======+=======+=======+======+=======+======+=======+======
{}   | ETL_t  | manua  | 2026-  | 2026-  | None   | None | 2026-  | manu | None   | queue
     | oll_d  | l__20  | 03-18  | 03-19  |        |      | 03-19  | al   |        | d
     | ata    | 26-03  |        |        |        |      |        |      |        |
     |        | -19T1  | 16:42  | 16:42  |        |      | 16:42  |      |        |
     |        | 6:42:  | :44+0  | :44+0  |        |      | :44+0  |      |        |
     |        | 44+00  | 0:00   | 0:00   |        |      | 0:00   |      |        |
     |        | :00    |        |        |        |      |        |      |        |
```

```
dag_id        | run_id        | state    | execution_date | start_date     | end_date
==============+===============+=========+================+================+================
ETL_toll_data | manual__2026- | success | 2026-03-19T16: | 2026-03-19T16 | 2026-03-19T16:4
              | 03-19T16:42:4 |         | 42:44+00:00    | :42:45.883756 | 3:03.874645+00:
              | 4+00:00       |         |                | +00:00         | 00
```